

■ B.1 Basic Objects

■ B.1.1 Expressions

Expressions are the main type of data in *Mathematica*.

Expressions can be written in the form $h[e_1, e_2, \dots]$. The object h is known generically as the *head* of the expression. The e_i are termed the *elements* of the expression. Both the head and the elements may themselves be expressions.

The *parts* of an expression can be referred to by numerical indices. The head has index 0; element e_i has index i . `expr[[i]]` gives the part of `expr` with index i . Negative indices count from the end.

`expr[[i1, i2, ...]]` gives the piece of `expr` found by successively extracting parts of subexpressions with indices $i_1, i_2, \dots$. If you think of expressions as trees, the indices specify which branch to take at each node as you descend from the root.

The pieces of an expression that are specified by giving a sequence of exactly n indices are defined to be at *level* n in the expression. You can use levels to determine the domain of application of functions like `Map`. Level 0 corresponds to the whole expression.

The *depth* of an expression is defined to be the maximum number of indices needed to specify any part of it. A negative level number $-n$ refers to all parts of an expression that have depth n .